



Looking Up: The Night Sky Over Billingborough

16 September – 14 October 2026

A practical guide to the most interesting astronomical events visible from Billingborough

The next four weeks bring increasingly long, dark nights and several worthwhile targets for amateur astronomers. There is a useful mixture of planetary observing, lunar events, a close Moon–Mars encounter and a meteor shower. The charts below are intended as simple visual guides to where to look from Billingborough.

At a glance

Date	Event	What to look for
18 September	Lunar X and V	Look at the Moon around 19:00 BST for the striking light-and-shadow effects.
23 September	Autumnal equinox	The Sun crosses the celestial equator; nights now lengthen rapidly.
26 September	Neptune at opposition + Harvest Moon	Neptune is visible all night, but the full Moon will make finding it harder.
27 September	Moon near Saturn	A bright Moon passes close to Saturn in the southern sky.
4 October	Saturn at opposition	Saturn is visible all night and reaches its brightest of the year.
5 October	Moon and Mars close together	Look southeast before sunrise; Jupiter is also visible lower down.
8–9 October	Draconid meteor shower	The radiant is high in the evening and moonlight is minimal.
10–12 October	Mars crosses M44	Mars passes through the Beehive Cluster in Cancer — a good binocular/telescope target.
13 October	Vesta at opposition	A telescopic asteroid target, around its brightest for the year.

The autumn nights begin

23 September – Autumnal equinox. At 01:04 BST on 23 September, the centre of the Sun crosses the celestial equator, marking the autumnal equinox in the Northern Hemisphere. From this point the nights become

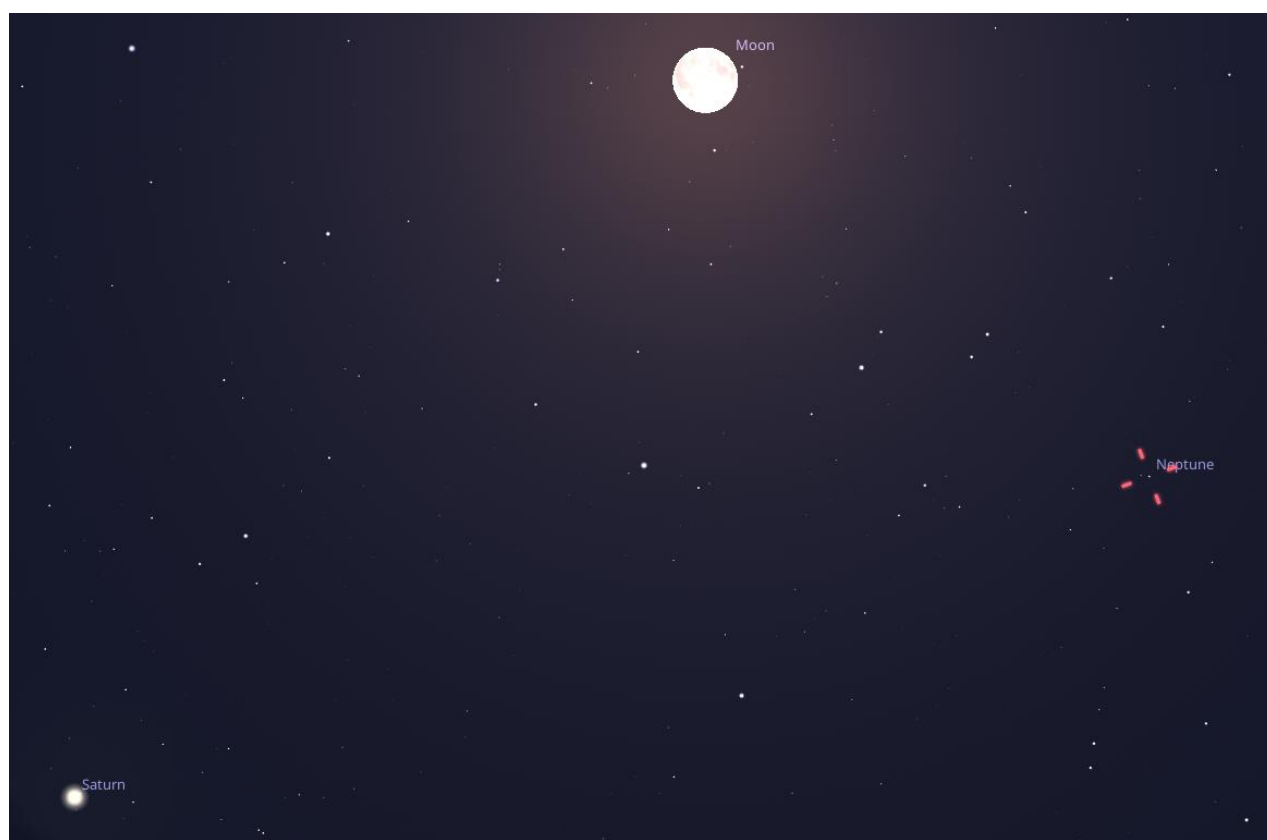
progressively longer, making the latter part of September and early October an increasingly rewarding time for observing.

18 September – Lunar X and V. Around 19:00 BST, the changing illumination of the Moon produces the well-known Lunar X and Lunar V clair-obscur effects. These are not permanent features of the lunar surface: they are created by the angle of sunlight falling across particular craters and ridges.

26 September – Neptune and the Harvest Moon

Neptune reaches opposition on 26 September, putting the distant planet on the opposite side of the sky from the Sun and making it visible throughout the night. Neptune is around magnitude 7.8, so binoculars or a telescope are needed; a telescope of roughly 150 mm aperture can show its tiny blue-hued disc.

The same night brings the 2026 Harvest Moon, the full Moon closest to the September equinox. This is a beautiful naked-eye event, but the bright Moon will also make Neptune and the surrounding faint stars considerably harder to pick out. For a darker-sky attempt at Neptune, the nights around the following new Moon will be more forgiving.



Looking southeast on 26 September: the Moon, Neptune and Saturn. Star chart generated using Stellarium.

4 October – Saturn at opposition

Saturn reaches opposition on 4 October. The ringed planet will be visible from around sunset through to sunrise and reaches its highest point in the southern sky around the middle of the night. This is one of the best times of the year to observe Saturn through a telescope, with its rings and brighter moons providing rewarding targets.

5 October – Moon, Mars and the Beehive Cluster

Before sunrise on 5 October, the waning Moon makes a close visit to Mars in the southeast. Jupiter is also prominent lower down in the same broad area of sky. Nearby lies M44, the Beehive Cluster, a large open cluster in Cancer that is particularly attractive through binoculars.

This is an especially useful observing opportunity because the Moon provides an obvious starting point. Once Mars has been located, the Beehive Cluster can be found nearby against the background stars.



The pre-dawn southeast sky on 5 October, showing the Moon, Mars, M44 (the Beehive Cluster) and Jupiter. Star chart generated using Stellarium.

8–9 October – The Draconid meteor shower

The Draconids are active from about 6–10 October, with the predicted maximum at approximately 01:00 UT on 9 October. Unlike many meteor showers, the Draconid radiant is already high in the evening, so there is no need to wait until the early hours to begin watching. The International Meteor Organization gives a typical rate of only around five meteors per hour, but the shower is historically capable of occasional unexpected outbursts.

The timing is particularly favourable in 2026 because the Moon is only a thin waning crescent around the peak and reaches new Moon on 10 October. For the best chance of seeing meteors, find a dark location with a broad view of the sky, allow your eyes 20–30 minutes to adapt to darkness, and look generally away from the radiant in Draco.

10–12 October – Mars meets the Beehive

Mars continues its journey across the background stars and passes through the Beehive Cluster, M44, between 10 and 12 October. This is a particularly nice target for binoculars or a wide-field telescope, because the orange-red planet will contrast with the cluster's many stars. The new Moon on 10 October also means very little moonlight will interfere with the view.

13 October – Vesta at opposition

The asteroid 4 Vesta reaches opposition on 13 October. Vesta is not a naked-eye target from ordinary suburban skies, but it can be followed with a telescope by comparing its position with a suitable star chart over successive nights. For observers who enjoy hunting down minor planets, this provides an additional challenge during an otherwise busy period.

A good four weeks for observing

For a general-purpose observing session, the period from the beginning of October through the Draconid peak is particularly attractive. Saturn is near opposition, Mars is moving through Cancer, the Moon becomes a useful guide to the morning planets, and the skies become increasingly dark as the Moon approaches its new phase. From Billingborough, a clear evening with a good southern and eastern horizon should provide plenty to explore.

Sources and checking

Billingborough Observatory

Event dates and observing information were checked against published 2026 observing guides. The two sky charts reproduced in this document were supplied by Billingborough Observatory and were generated using Stellarium, the free open-source planetarium software: stellarium.org.